Environment International 191 (2024) 108992

Contents lists available at ScienceDirect

Environment International

journal homepage: www.elsevier.com/locate/envint

Full length article

Combining Google traffic map with deep learning model to predict street-level traffic-related air pollutants in a complex urban environment

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ARTICLE INFO

Handling Editor: Dr. Xavier Querol

Keywords:
Machine learning
Deep learning
Mobile measurement
Street-level pollution
Air quality sensor
Crowd-sourced

ABSTRACT

Background: Traffic-related air pollution (TRAP) is a major contributor to urban pollution and varies sharply at the street level, posing a challenge for air quality modeling. Traditional land use regression models combined with data from fixed monitoring stations may be unable to predict and characterize fine-scale TRAP, especially in complex urban environments influenced by various features. This study aims to estimate fine-scale (50 m) concentrations of nitrogen oxides (NO and NO_2) in Hong Kong using a deep learning (DL) structured model.
Methods: We collected data from mobile air quality sensors on buses and crowd-sourced Google real-time traffic status as a proxy for real-time traffic emissions. Our DL model was compared with existing machine learning models to assess performance improvements. Using an interpretable machine learning method, we hierarchically evaluated the global, local, and interaction effects for different features.
Results: Our DL model outperformed existing machine learning models, achieving R^2 values of 0.72 for NO and 0.69 for NO_2. The incorporation of traffic status as a key predictor improved model performance by 9% to 17%. The interpretable machine learning method revealed the importance of traffic-related features and their pairwise interactions.
Conclusion: The results indicate that traffic-related features significantly contribute to TRAP and provide insights and guidance for urban planning. By incorporating crowd-sourced Google traffic information, we assessed traffic abatement scenarios that could inform targeted strategies for improving urban air quality.

1. Introduction

Traffic-related air pollution (TRAP) produced by the emission of motor vehicles is a major contributor to urban air pollution (Grange et al., 2017; Guarnieri and Balmes, 2014; Harrison et al., 2021), including nitric oxide (NO), nitrogen dioxide (NO_2), black carbon (BC), VOCs (Volatile Organic Compounds) and particulate matter. Evidence is mounting that long-term exposure to TRAP is linked to several adverse health outcomes, such as cardiovascular disease, respiratory symptoms and asthma (Ji et al., 2024; Zhu et al., 2023), premature mortality and dementia (Chen et al., 2017; HEI, 2010; World Health Organization, 2016), especially for vulnerable groups living in proximity to or at busy streets (Hoek et al., 2002; Kim et al., 2004). In urban areas, most people who live within 500 m away from major roads suffer from the health effects of long-term or short-term exposure to air pollution (Delfino et al., 2008; Su et al., 2015). Governments and environmental agencies deploy in-situ regulatory stations to monitor the air quality and establish regulations with guidelines or standards. Limited by the high cost and maintenance, conventional fixed-site monitoring stations are often sparsely distributed that cannot capture spatial variations of pollutants at fine scales(Kumar et al., 2015).

Mobile measurements provide an alternate way to capture pollutant

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https://doi.org/10.1016/j.envint.2024.108992
Received 28 March 2024; Received in revised form 26 August 2024; Accepted 29 August 2024
Available online 6 September 2024

variations at sub-kilometer scales in urban environments(Apte et al., 2017; Hatzopoulou et al., 2012). To estimate fine-grained spatial variation of pollutants concentrations, several mobile sampling strategies coupled with empirical land use regression (LUR) models have been developed and demonstrated the capacity to quantify human exposure (C. Simon et al., 2018; Dalgleish et al., 2018; Deville Cavellin et al., 2015; Hankey et al., 2019; Hankey and D. Marshall, 2015; Shi et al., 2016). Recently, rapid advances in internet of things (IOT) technologies and low-cost air quality sensors have extended the scope of air quality measurement in multiple settings (Kumar et al., 2015; Mead et al., 2013; Thompson, 2016). Data quality from these sensor-based measurements is enhanced by evolving calibration paradigms and hardware designs (Baron and Saffell, 2017; Kumar et al., 2015; Morawska et al., 2018). These low-cost air quality sensors can be deployed in varied scenarios to expand existing monitoring networks or mobile campaigns in complex urban environments (Hatzopoulou et al., 2017; Morawska et al., 2018).

The LUR model aims to estimate air pollutant concentrations and assess individual exposure levels in epidemiological studies (Briggs, 2006; Hoek et al., 2008a). It utilizes geographic covariates as spatial predictors, e.g., land use types, traffic and transport infrastructures (Dalgleish et al., 2018; Hoek et al., 2008a). Linear relationships in the LUR model may overlook complex nonlinearities between air pollutant concentrations and predictors. Machine Learning (ML) based air quality models, such as artificial neural network (ANN), random forest (RF), and XGBoost, offer improved accuracy and resistance to multicollinearity compared to linear regression models (Di et al., 2016; Ding et al., 2021; Hu et al., 2017; Li et al., 2017; Xiao et al., 2018; Zhou et al., 2021; Ding et al., 2021; Xiao et al., 2018; Zhou et al., 2021). On the other hand, deep learning (DL) models have gained attention for extracting complex features in hierarchical representations for air pollution prediction (Lyu et al., 2019; Pak et al., 2020; Yan et al., 2020). Meanwhile, ensemble strategies integrate multiple base models in air pollution modeling, combining their advantages for improved predictions (Di et al., 2019; Xiao et al., 2018). Still, the complexity improves ML model's performance at the cost of the interpretability, to open the "black box" of the ML model and make it understandable is necessary to extend the insight of feature effects and interactions (Molnar et al., 2020; Reichstein et al., 2019; Zhong et al., 2021).

Traditional tabular predictors have limitations in representing fine-scale spatial variations of traffic emissions and dynamic transportation developments. Nontabular data enriched with detailed street-level air quality information can greatly contribute to air quality assessments when synchronized with mobile sensor measurements (Wang et al., 2023). The hybrid models incorporate outputs from dispersion as input for traffic emission simulation or as new data sources which enhanced the spatial predictability of the models (Gerges et al., 2024; Lloyd et al., 2023). Google Maps provides an array of non-tabular data, accessible as graphical information, while Google Street View (GSV) amasses a vast collection of panoramic imagery (Gong et al., 2018). Combined with deep learning models, GSV has been widely used to quantify urban characteristics and predictors (Gong et al., 2018; Hu et al., 2020; Li and Ratti, 2019). Meanwhile, Google Maps has used "crowd-sensed" data from phone users to analyze and predict real-time traffic congestion conditions (Google Maps, 2009; Nair et al., 2019). Compared with static traffic data, real-time traffic congestion status can supply more dynamic and up-to-date traffic information that is associated with transient vehicle emissions and can better indicate short-lived pollutants, such as NO_x, UFP and BC (Hilpert et al., 2019). Quantify and understand congestion-induced TRAP enables policymakers to make more informed decisions and targeting effective interventions to reduce exposure levels (Gately et al., 2017). Nevertheless, a robust modeling framework that integrates traffic congestion data and providing interpretable results in complex urban environments yet to be explored.

In this study, we employed low-cost sensors on fixed-route buses to measure NO and NO_2 across varied road environments in Hong Kong and built high performance model to estimate street-level TRAP. We introduced Google traffic congestion status as a predictor to estimate TRAP in the complex urban environment. Furthermore, we use model-agnostic strategies to quantify feature contributions and get a thorough understanding of the complex interactions between different variables that influence on TRAP. Finally, by evaluating the congestion influence on street-level pollution, we highlighted the importance of traffic improvement on TRAP abatement in key localized roads.

2. Materials and methods

2.1. Study area and air pollution measurement

Hong Kong's urban area is among the densest in the world's megacities. Over 7.5 million residents inhabit approximately 262 km^2 of developed land. Hong Kong has over 10,000 high-rise buildings (>40 m) (Emporis, 2018) and residential densities exceeding 110,000 people/km^2 (Figure S1). The study area encompasses Kowloon Peninsula and Hong Kong Island's ultra-dense urban buildings and road networks (Figure S2). In this region, the high building height-to-street width ratio creates deep street canyons. Narrow streets often collocated with high level of traffic congestion, exacerbating air pollution further. The mobile sensor measurements were deployed on double-deck buses that travel along fixed routes to capture repeat measurements during bus schedule hours. The monitoring campaign was conducted over the course of two months in 2017, specifically in April and July, aligning with the spring and summer seasons respectively in Hong Kong. This seasonal timing was chosen to assess environmental variations across different climatic conditions. NO and NO_2 were measured by electrochemical gas sensors with a sampling rate of 5 s. Details on the platform, sensor calibration, and data quality control are in our previous work (Wei et al., 2021) and Text S1. In the study area, ten selected bus routes covered multiple road types and environments. Figure S3 represents the mobile measurement routes and the percentage of road types in the study area. The Hong Kong road network was segmented into equal lengths, and all measurements were allocated to the nearest road segments. The mobile measurements campaign covered 130 km, and a total length of 100 km was selected with a repeat measurement number of more than 20 times (about 2000 road segments). We chose 50 m resolution to balance the representativeness of the measurement and route repeat numbers (Figure S4). All measurements were allocated to road segments for further model building(Dalgleish et al., 2018; J. Miller et al., 2020). First, we assigned the mean concentration for each session of unique drive pass to its corresponding road segment. Then, we calculated the median of these drive pass mean concentration values for each road segment. This median concentration was used to represent the typical level of pollution on that road segment. To ensure the representativeness of measurements, we excluded road segments that were visited fewer than 20 times (Hatzopoulou et al., 2017). As a comparison, we also used grid based geo-processing method to reconstruct measurements and variables (Shi et al., 2016). Meanwhile, we followed Apte's steps that used multiplicative time-of-day factor to correct diurnal background variation (Apte et al., 2017). The adjustment factor was based on Hong Kong Environmental Protection Department (HKEPD) roadside air quality station. The influence of meteorological conditions was assessed (wind speed and direction) for individual routes at the microscale (50 m). The result revealed non-significant p-values (>0.05) for most routes, indicating that wind speed and direction did not exert a dominant influence on NO and NO_2 concentrations compared to traffic-related features. Detailed statistical results are provided in the supplementary materials (Table S1).

2.2. Geo-Predictor variables and processing

Based on previous LUR model building (Hoek et al., 2008b; Su et al., 2009), we collected four types of traditional spatial predictors to characterize traffic-related emission intensity: transport infrastructure, land use types, and population. The land use variables cannot fully represent Hong Kong's high-density and variation of urban building morphology compared with low-density urban environment. We followed our previous work which used urban morphology features to characterize Hong Kong's unique urban environment (Shi et al., 2016). All variables and summaries are listed in Table S2 with detailed information. We used road segment centroid variables and different buffer sizes to generate candidate variables (353 in total) for models.

2.3. Google map predictors

Google maps platform (GMP) allows users to use an application program interface (API) to connect the platform and extract information. We selected two modes of GMP free data as predictors.

2.3.1. Google street view factor

For each road segment, we collected centroid panorama images via API key to access corresponding GSV ID and download each tile image. Afterward, all tiles were combined to generate a panorama image. The GSV images collected in the same year (2017) were collected to match the mobile measurement. We used a scene parsing based deep learning model to parse an image into varied semantic categories and extract street-level features (Text S3, Figure S5) (Zhao et al., 2017). Considering Hong Kong's unique urban morphology with narrow streets forming a street canyon environment, we reproject the segregated panorama images onto a circular plane and generate a fisheye image based photographic method instead of the plane percentage calculation (Li et al., 2018). Detailed information and calculation equations are elaborated in Text S3. The view factors (VFs) of the sky (SVF), building (BVF) and trees (TVF) were selected as candidate predictors covering approximately 95 % of street view in the study area. We averaged the VFs of two years (2017 and 2018) to decrease the influence of sunshine on segmentation in the high-rising areas and used the buffered areas of VFs as candidate predictors.

2.3.2. Real-Time Google traffic index (GTI)

Google Maps display real-time traffic status with a color-coded system that overlays on the base map with multiple map elements. We used GMP to create a traffic status-only layer via API, extracting traffic data from Google Maps. The redesigned map is saved as time-labeled images during mobile measurements. An image processing and identification algorithm was developed to pick up traffic status (Figure S6). We define Google traffic index (GTI) ranged from 1 to 4 to indicate the congestion status (Table S3). Finally, each road segment was assigned with GTI that correspond to the mobile measurements time. We sampled traffic map at a 3 min interval and synchronized with the mobile measurement campaign time. The detailed steps for traffic status image identification and road network map fusion are explained in Text S4. The on-road traffic congestion status GTI and buffered GTI which represent neighborhood congestion status influences were calculated as candidate predictors. Meanwhile, the GTI was also integrated with traffic volume to represent the influence of vehicle fleet traffic status on-road pollution. Accordingly, the traffic index in synchronization with bus GPS speed was converted into real vehicle fleet speed for different types of roads (Table S4).

2.4. Model building

We built four types of prediction models to assess their performance: the basic LUR model was developed as a benchmark model and feature selection, machine learning models, multi-layered deep learning models and an ensemble model which utilized different single models as base estimators (Fig. 1). It should be noted that the fundamental unit of analysis and prediction is the road segment at a resolution of 50 m,

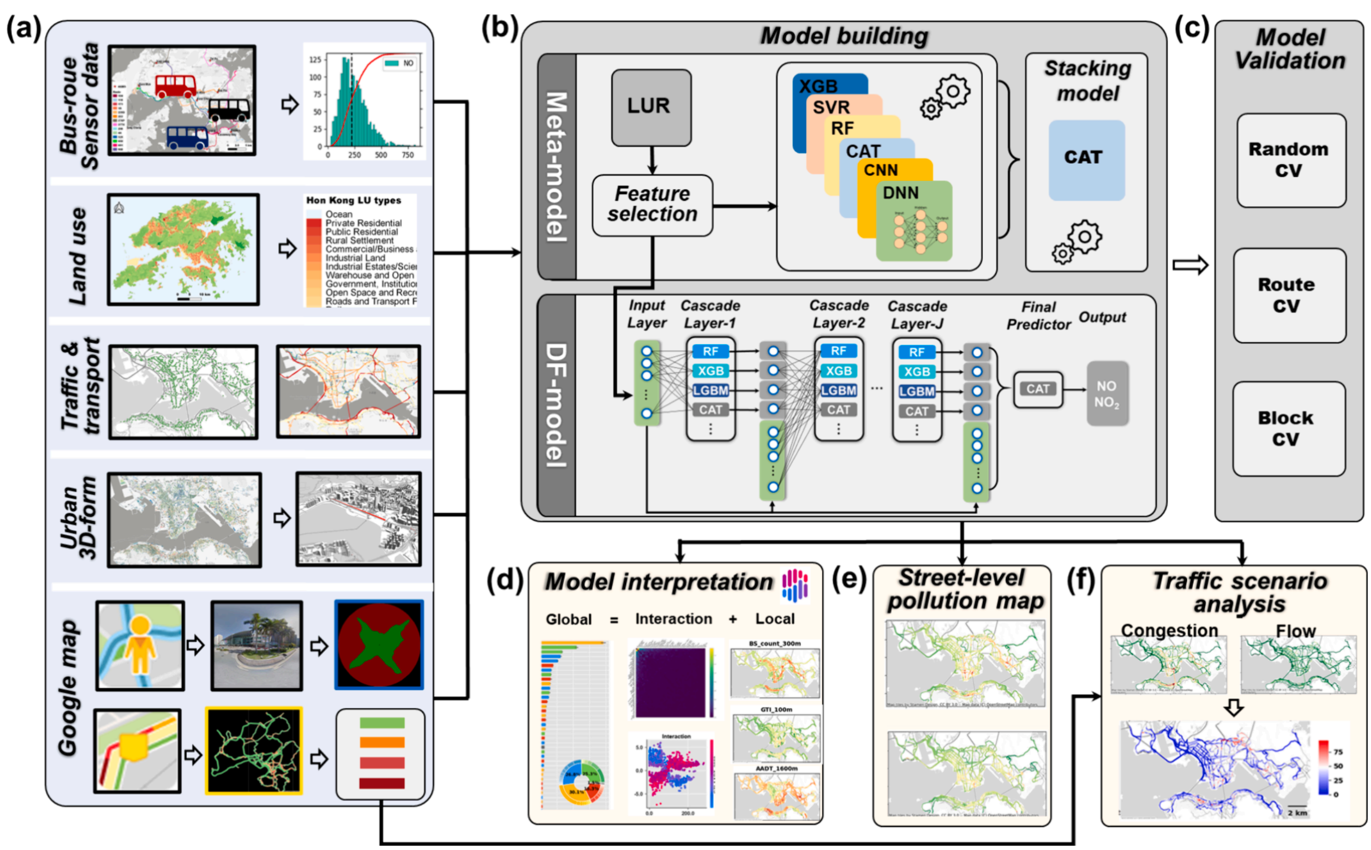

Fig. 1. The framework of the model building and applications in this study. (a) Data parts, (b) model building, (c) model CV strategies, (d) model interpretation, (e) prediction and (f) scenario analysis.

tailored for assessing the on-road ambient environment.

2.4.1. LUR model building

We performed a two-step strategy on the basic LUR model building. Firstly, we used a distance-decay regression selection strategy (ADDRESS)(Shi et al., 2016; Su et al., 2009) to screen candidate variables (both on-road and buffered variables) as model inputs for the second step and other models. For each type of variable, only variables with the highest correlation coefficient with the target pollutant were selected as model input. This step can avoid multicollinearity between different buffer size variables(Shi et al., 2016). Secondly, a forward selection procedure was conducted by starting with the variable that had the highest adjusted R^2. The variables that increased the model adjusted R^2 by more than 0.01 and reduced the model residual error were retained. The model building process was terminated when a variable is not significant ($p < 0.1$) to the model or the variance inflation factor (VIF) exceeded 5 (Hankey et al., 2017).

2.4.2. Machine learning (ML) models

We applied five supervised ML models to estimate street-level air pollution and compare their performance: support vector regression (SVR), random forest (RF), light gradient boosting Machine (LGBM), extreme gradient boosting (XGB), CatBoost (CAT). Detailed explanations of the stacking model building and structure can be found in Text S5 and Figure S8. RF is a bagging method which integrating multiple decision trees with each node of the tree is split according to the best of a subset of randomly predictors. Unlike RF, GBDT used boosting method by sequentially connected decision trees to obtain a strong learner. XGB was developed based on GBDT and evolved by adding regularization term which reduced model complexity and accelerated solution process. CAT is also a gradient boosting on decision trees-based framework that utilizes a boosting method that is well-suited for classification or regression tasks that involve categorical and heterogeneous datasets (Hancock and Khoshgoftaar, 2020). The development of CAT occurred more recently compared to other models, and it has demonstrated a growing potential in recent times (Ding et al., 2021; Zhou et al., 2021). For all ML models training stages, the hyperparameters were tuned with a grid-search method and are listed in Table S5. We employed a stacking method that learns from the prediction results of several base ML models and used a *meta*-model for the final prediction. The stacking strategy could utilize advantage of different models performance that varied at spatial and outperformance than single model. The base ML models were developed and CAT was chosen for the *meta*-model. The stacking method can be extended to multiple layers, thus we built a 2-layer and 3-layer structured model to evaluate model performances.

2.4.3. Deep learning (DL) models

We used a deep neural network (DNN) model and a modified convolutional neural networks (CNN) model, ResNet, for our regression prediction task. ResNet often contains deep hidden layers in the network and shows high performance for image recognition works. We modified ResNet model for regression task by replaced convolutional layers and pooling layers with fully connected layers in the residual block (Figure S9) (Chen et al., 2020). The DL models were optimized with searching layers and hyperparameters (Table S5).

Deep forest (DF) model is a deep learning structured model and used multiple ensemble base models to replace neurons in the hidden layer (Zhou and Feng, 2019), in other words, DF combines DL structure and stacking strategy. The base learners contain RF and extreme trees (ET) which constituted hidden layers. The ADDRESS selected features were as input for best models (ranking by performance) in the cascade layer, the prediction result was concatenated with raw input features as augment features and as input for the next layer (Fig. 1(b)). DF model training process will terminate if there is no significant performance gain. For each hidden layer, a cross validation (CV) was carried out to avoid overfitting. Finally, a *meta*-predictor is connected to the augmented outputs by the last hidden layer which produce the final prediction results. In this study, we used DF model structure and built-in base models were replaced with optimized base models developed in Section 2.4.2 to serve as estimators and predictors.

2.5. Model-Agnostic interpretation

We employed a model-agnostic (compared to model-specific) interpretation method for "black-box" ML and DL models (Fig. 1(d)), which enables us to compare multiple ML models flexibly (Molnar, 2019; Molnar et al., 2020). The SHapley Additive exPlainations (SHAP) is a model-agnostic interpretation technique that calculates feature importance using Shapley values, which are grounded in game theory (Lundberg and Lee, 2017). SHAP values provide a unified approach for explaining model results through local explanations based on the concept of additive feature attribution. By summing the marginal contributions of all instances, SHAP values can evaluate global feature importance. Unlike built-in feature importance for special models which only show a global feature characteristic, SHAP values provide local interpretability, enabling a detailed examination of each individual instance. Additionally, SHAP can also generalize feature interaction effects instead of assuming that features are mutually independent. Text S6 gives a detailed description of SHAP values of global and local interpretations and interaction effects calculation.

2.6. Model evaluation and predictions

The model performance was evaluated by adjusted R^2, root-mean-square-error (RMSE), and 5-fold cross-validation (CV). Apart from random split CV, we designed route-based CV and spatial block CV evaluation strategies (Fig. 1(c)) to assess model prediction robustness (Text S7, Figure S9). For the segment-based prediction, we calculated the required predictors for each road segment centroid to estimate NO and NO_2 concentrations with the optimal DF model (Fig. 1(e)).

Further, leveraging the Google traffic data, we designed a counterfactual scenario (Fig. 1(f)) to probe the impact of traffic status transition from real-world to congestion-free status on road pollution. Two types of strategies were considered: first, a predictive comparative analysis (or what-if) analysis to simulate potential effects of transitioning from congestion to a free status, and second, a causal analysis employing a nonlinear Double Machine Learning (DML) to estimate average causal effects of these changes, more details can be found in Text S8.

3. Results

3.1. Mobile measurements and model performance

The statistic of mobile measurements and corresponding co-variables are shown in Figure S10. The median concentrations of on-road NO, NO_2 are 250 ppb and 40 ppb, respectively, which present skewed distributions indicating high concentration segments and spatial heterogeneity. The on-road TRAP revealed high levels of pollution and spatial variations (Figure S11), similar findings could be found in previous studies that were conducted in Hong Kong and many other Chinese megacities (Lee et al., 2017; Li et al., 2019). Such ambient pollution levels were manifold of those from mobile campaigns deployed in the U.S and European countries (Apte et al., 2017).

The performance of four category models was compared in Fig. 2(a, b), and the RMSE and MAPE results are detailed in Table S7. As a benchmark, the LUR model showed the lowest performance as expected with 5-fold CV R^2 of 0.41 and 0.36 for NO and NO_2, which indicates that the linear model can hardly explain the pollution variation in the complex urban environment (Lee et al., 2017; Miskell et al., 2015). The ML models present better performance than LUR model and vary with the mechanism. Generally, the classification tree-based models tend to show close performance and prediction results (Figure S12, S13), such as

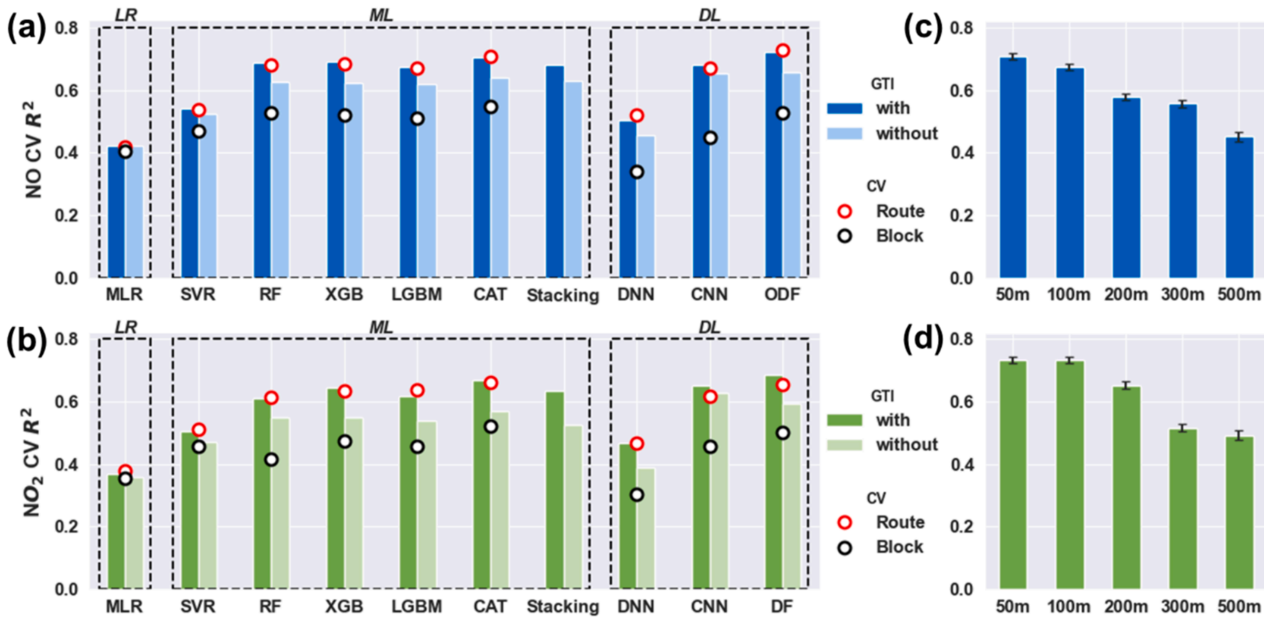

Fig. 2. Model performance of R^2 for NO (a) and NO_2 (b). The red dots are bus-route CV, and black dots are spatial block CV. The light bars are the model performance without GTI related variables (the stacking model only supported random CV). The spatial resolution performance for NO (c) and NO_2 (d), the error bar represents 100 times randomly CV results. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

RF and XGB with delta R^2 within 0.02. For all base models, the CAT presents superior performance and was chosen as the final estimator for DF model. However, the stacking method (two layers stacking model) merely presents moderate performance instead of superior performance than base models as some studies found (Kerckhoffs et al., 2019; Liu et al., 2021; Shtein et al., 2019). When the stacking layer increased from 2 layers to 3 layers, the model performance even decreased at the cost of R^2 reduction of 0.04 for NO and 0.07 for NO_2 (Figure S14, Table S9) which indicates the complexity enhanced overfitting. For DL models, DNN presents mediocre performance with slight improvements compared with LUR model, while the CNN could extract hierarchical features in convolutional layers that improved performance compared to DNN. The DF model shows the highest performance with CV R^2 of 0.72 and 0.68 for NO and NO_2. Different from the stacking model, the DF model performance was constrained after layer-by-layer processing. The results suggest that, in addition to improving prediction performance, the DF model addressed both generalization and overfitting. To extensively assess the model generalization ability, we also evaluated route-based CV and spatial-block CV strategies. The route CV presents similar performance with random CV for all models. The model spatial extrapolation was expectedly undermined about 0.15–0.17 which is consistent with previous studies (Lu et al., 2021). The comparison and DF model predictions could be seen in Figure S15.

As a comparison of spatial variables construction schemes, we also evaluated segments-based prediction and grid-based prediction using the DF model at the same spatial resolution (Figure S16) which showed a strong consistency (*Pearson* $r > 0.85$). On this basis, we investigated the influence of spatial resolution on model performance using grid-based variables ranging from 50 m to 500 m ((Fig. 2(c,d)). The fine-scale (50 m) grid-based model performance is close to that of the segment-based model. Both NO and NO_2 model performance present decreasing trend with the upscaling of resolution (Figure S17), suggesting that coarse resolution could weaken the model prediction ability and result in a loss of spatial details. Correspondingly, NO was more sensitive to spatial resolution than NO_2 indicating the hyperlocal variation effect (Chambliss et al., 2021).

3.2. Model interpretation

The global feature importance is presented in Fig. 3(a,b). Transport infrastructure and traffic-related features contribute the most, accounting for more than 52 % and 64 % for NO and NO_2. Notably, our new traffic status variables rank top in feature importance, the on-road (GTI) and neighborhood (within 200 m) traffic status can capture instant localized traffic activities and are key contributors of traffic emissions. The introduction of GTI substantially improved model prediction, increasing R^2 by 0.05–0.1 for non-linear models (Table S8). Concurrently, bus stop density (BS_count), road length density and traffic volumes (AADT) influence pollutant emission distributions. Unlike localized traffic status, commercial areas exert a wider influence on pollution levels by directing transportation activities at a larger scale. Aside from emission intensity related variables, urban form is another non-negligible factor that influences pollutant dispersion or accumulation conditions (Shi et al., 2016). The shared top ranking features and buffer radius suggest similarities in the driving factors and influence range for NO and NO_2 (Lee et al., 2017). Comparison of feature importance between base models (Figure S18, S19) indicates that tree-based models exhibit similar feature rankings to the overall model performance. Further, the SHAP summary plot provides evidence of variable influence directions on model prediction for individual instances (Figure S20). Overall, traffic emission related variables exhibit positive contributions, while variables related to dispersion status, e.g., the SVF that indicates openness of urban form, present negative effects.

Leveraged by SHAP dependence plot, we can better understand the relationships between pollutants and variations in each feature. Fig. 3(d) illustrates the impact of bus stop density on the predictions of each measurement and presents positive contributions to pollutants with a nonlinear trend. Generally, most features exhibit non-monotonic trends suggesting that the influence of pollutants varies regionally (Figures S21-24, Text S7). This could explain the poor performance of the linear regression based LUR model in predicting pollutant variations.

Except for univariate analysis, we investigated coupling intensities of main features with SHAP interaction values. Fig. 3(c) presents feature interaction values for NO (Figure S25 for NO_2). To summarize, on-road pollution is driven by a combination of traffic status, transportation, land use and urban form. The ranking of feature interaction values

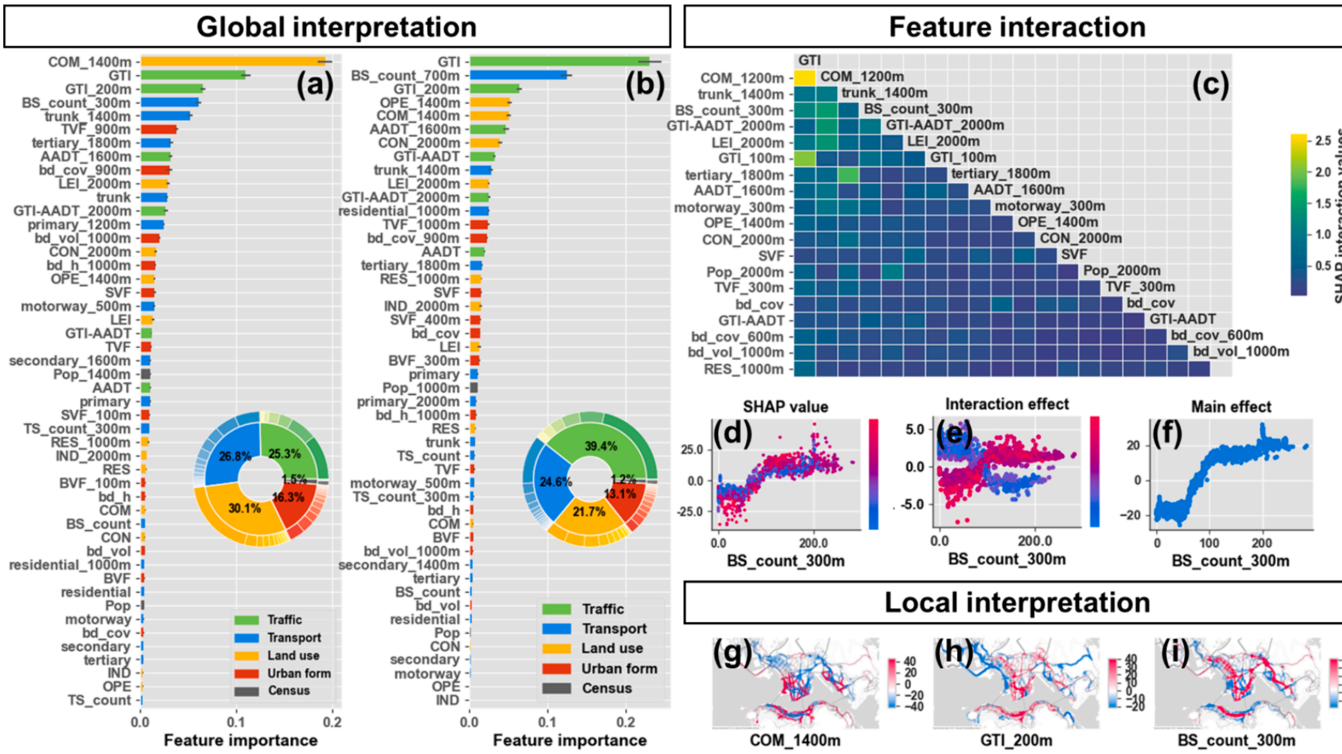

Fig. 3. SHAP explained global feature importance for NO (a) and NO_2 (b). The inset PieDonut contains categorized features (inner ring) and single variable contributions (out ring). The summary plots are presented in Figure S19. (c) The feature interaction values of the top 20 features for NO, the y-axis is interaction features which is ranked by mean interaction values. (d) SHAP values dependence plot for bus station count, the vertical dispersion dots indicate magnitude of interaction effects. The dots (measurements) are colored by traffic status and traffic volume (GTI-AADT), with red indicating congestion and high traffic values and blue indicating inverse. Vertical dispersions of colors depict the magnitude of interaction effect. (e) The separated interaction SHAP values between bus stop count and traffic status with volume. (f) The main effect SHAP values for bus stop density contribution to pollutant prediction. The spatial distribution of SHAP values for three top features for NO prediction: commercial land use (g), traffic status (h) and bus station distribution (i). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

closely aligns with individual feature importance, suggesting that on-road pollutions are not solely influenced by dominant features, but rather by feature interactions as well. The partial dependent plot in Fig. 3(d) shows the combined effect contributions to SHAP values, the vertical dispersion of SHAP values for a certain of bus stop density (BS_count) value is due to the interaction effect with congestion and traffic volume (GTI_AADT). By decoupling the interaction effect from the feature contributions, we obtained main effect of BS_count in Fig. 3 (e) and feature interaction SHAP values between BS_count with GTI_AADT in Fig. 3(f). The main effect illustrates positive contributions of bus stop density to SHAP values. The interaction presents a positive contribution with the increase of BS_count and higher GTI_AADT (red dots), which indicates high bus stop density (>100 for 300 m buffer area) coupled with traffic congestion enhanced pollution level collectively. An inverse case could be observed between urban form factor (SVF) and traffic status in Figure S25. The negative interaction effect between increased SVF values and high GTI indicates street openness could mitigate congestion induced pollution. The turning point occurred at higher SVF values (>0.75). A summary of the interactions between four types of key features is shown in Figure S26 and explained in Text S6.

The SHAP interpretation map emphasizes the importance of local (each road segment) contributions and indicates regions where variables have positive or negative associations with the TRAP. We estimated the contributions of the three most important features SHAP values for NO in Fig. 3(g,h,i). High-density commercial area (COM_1400m) is mainly along the periphery of Hong Kong Island and the tip area of the Kowloon Peninsula, resulting in the highest positive SHAP values. The traffic status (GTI_200m) emphasizes that congestion-related contributions are concentrated along certain roads. The SHAP value for bus stop exhibits an aggregated spatial pattern that indicates the potential for TRAP accumulation. Conversely, a lower station density design could help mitigate TRAP with negative SHAP values. Figure S27, S28 and Text S6 give the summary of main feature SHAP maps for NO and NO_2. From a comprehensive perspective, the sum of individual SHAP values forms the final prediction, with pollution variation explained by discrepancies in feature compositions.

3.3. Predictions and traffic congestion Impacts

The on-road TRAP estimations are illustrated in Fig. 4(a,b). Dissimilar to the highway-dominated elevated pattern (Apte et al., 2017; J. Miller et al., 2020), NO and NO_2 concentrations in Hong Kong exhibit centralized spatial patterns (projected profiles), with major road hotspots mainly identified in core urban areas (Che et al., 2023; Lee et al., 2017). Generally, heavy pollution is distributed at Central, Causeway Bay areas which are Hong Kong AQMS monitoring areas, as well as in Kwun Tong area and Hung Hom Tunnel entrance where there is heavy traffic activity. High concentrations are also observed along Nathan Road, which is surrounded by high-density buildings. Grid-based predictions also present similar distributions (Figure S17).

The predictions of congestion-free status are shown in Fig. 4(c,d). The pollution abatement was calculated based on the predictive comparative analysis in Fig. 4(e,f). The reductions presented a heterogenous and aggregating pattern as pollution distributions. Globally, pollution reductions in most of road segments are limited (below 10 %) that can hardly benefit from the eliminated congestion. Whereas individual road segments could benefit from higher reductions with more

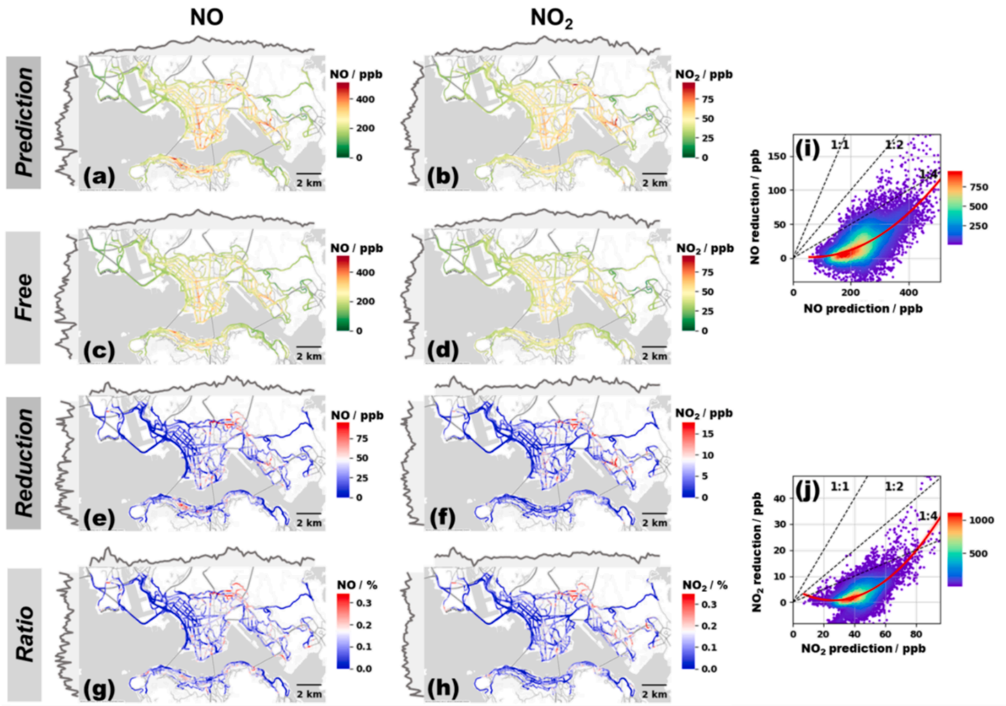

Fig. 4. The prediction of on-road concentrations of current status (a, b) and free status (c, d) for NO and NO_2 in the study area. The estimated reductions (e,f) are due to the improved traffic status and the ratio of the reduction compared to base prediction (g,h). The correlations between reduction and base prediction that marked with point density (i,j). Dash lines represent 1:4 (25% reduction), 1:2 (50% reduction) and 1:1 (100% reduction) relationship. The red lines are quadratic fit curves of reduction corresponding to predicted pollution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

than 25 % or even as high as 50 % (Fig. 4(i,j)). Interestingly, the reduction hotspots coincide with pollution hotspot areas and enhanced reductions with the increase in pollution levels (Fig. 4(g,h)). In other words, the congestion induced pollution plays a key role in heavily polluted road segments, and the elimination of heavily congested areas could gain augmented benefits (Table S10). The spatial co-occurrence of congestion-induced air pollutants indicates analogous emission patterns for NO and NO_2. Additionally, they also exhibit diurnal variation markedly, with peak levels observed during rush hours, and display consistent spatial patterns (Figure S29, S30). Similar rates of reduction were observed in the causal analysis and consistent with the predictive analysis (Table S10, Figure S31). On average, the reductions estimated from predictive comparative analysis are 7.7 % and 5.1 % for NO and NO_2, while 6.6 % and 6.7 % from causal analysis. Nevertheless, some variations in spatial distribution could be observed (Figure S32). This suggests that while there is a correlation between traffic status and pollution levels, the causal analysis also plays a more nuanced role, within which specific traffic conditions may exert a direct influence on the degree and distribution of TRAPs.

4. Discussion

4.1. Street-Level air quality model Development, interpretation and predictions

The base LUR model displayed the lowest performance and was poor than other cities (Table S11), indicating that the linear relationship constrains the description of pollutant variation, particularly in Hong Kong's intricate urban setting (Lee et al., 2017; Miskell et al., 2015; Ren et al., 2020). Lee et al. (Lee et al., 2017) reported lower LUR model performances for roadside sampling in Hong Kong, with CV R^2 0.28 and 0.39 for NO and NO_2. The ML model performance improved significantly through non-linear regressions. As a comparison, our base model prediction presents worthy performance among existing models (Table S11). Nevertheless, the main target of this study is not simply comparing with previous works or intercomparison of existing models, since ML models performance is context-based and may vary with urban environments and measurements (Kerckhoffs et al., 2019). Instead, improving existing model performance is prioritized in this work. Given that the model complexity is an indispensable condition for improving model performance (Poggio et al., 2020; Reichstein et al., 2019), we constructed the model in two dimensions: 1) model diversity by integrating various models; and 2) model structure by increasing the model depth. The stacking model followed this path and exhibited good performance in a number of studies (Lim et al., 2019; Liu et al., 2021; Shtein et al., 2019). However, it merely reached a moderate performance among base models and decreased with layers indicates overfitting constrained model performance. In other words, the stacking model induction ability was enhanced with depth, while the measurement noises or specificities were also learned in the training process, undermining model generalization ability. Our reconstructed DF model performed superior to other models by integrating diversiform models with a flexible structure, which adjusts complexity (layer number) adaptively instead of hyper-parameter tuning (Zhou and Feng, 2019). Furthermore, the DF model reached stable performance after layer-by-layer processing rather than relying on backpropagation or gradient descent for generalization in NN-structured DL models which are sensitive to model structure and hyperparameters.

The model resolution represents its ability to capture pollution hotspots and fine-scale spatial variations, which are crucial for accurately estimating personal exposure in environmental justice studies (Gardner-Frolick et al., 2022). The decreasing trend in model performance associated with upscaled measurements and variables implies that the prediction capacity to capture hyperlocal pollution variations at fine scales diminishes. This can be attributed to two main factors: 1) street-level TRAP measurements display localized variations but are detrended at coarser-scale backgrounds; 2) fine-scale variables like GTI better reflect localized pollution variations from direct emissions (Tang et al., 2013), which weakens at coarser scales. Our findings underscore the need for both fine-scale mobile campaigns and variables in street-level air quality model construction, as the former captures hyperlocal TRAP variations and the latter provides information directly related to street-level traffic emissions.

Besides enhancing prediction performance, comprehending feature-driven complexity and interactions is vital for demystifying ML models' "black box" nature. Globally, transport infrastructure and traffic-related variables contribute most to TRAP at street-level environments (Hatzopoulou et al., 2017; Lee et al., 2017; Miskell et al., 2015). The newly introduced GTI ranks the top predictor and presents substantial improvements in TRAP estimations. It could capture fine-scale traffic pattern variations related to localized vehicle fleet activities (Figure S33), while buffered GTI variables emphasize the neighborhood impact of congestion-induced traffic emissions. Crowdsourced traffic data offers near real-time information compared to traditional geographic variables, enabling rapid response to urban infrastructure construction. Combining open access data enables more mobile campaign deployment opportunities in multiple cities, facilitating intercity model building and street-level traffic pollution estimation in information-limited areas (Zalzal et al., 2019). Furthermore, real-time traffic data can provide localized road statuses like blockages or accidents that directly impact traffic emissions and serve as a crucial driver for spatiotemporal air quality model building and prediction. Other traffic-related variables, confirmed in prior studies (Jain et al., 2021; Lee et al., 2017), include road length as a crucial model prediction feature, while dominant road types may differ with surroundings. Bus stop density also serves as a significant estimator. Given public transport's vital role in Hong Kong, bus station arrangements can affect pollutant emissions in central urban or CBD areas locally (Ma et al., 2019; Miskell et al., 2015; Weissert et al., 2018). The introduction of VFs contributes to explain pollutant dispersion, especially in street-level scenes, and serves as attainable factors for keeping up with mobile measurements. Specifically, the SVF exhibits a negative effect, signifying the influence on pollutant dispersion status.

Feature interaction is rarely examined in urban air quality models that can intensify or mitigate pollution (Kerckhoffs et al., 2019), highlighting synergistic rather than univariate influences. These feature interactions varied with informative characteristics for implementing effective pollution control strategies. The neglection of features interplay effect may partially account for the linear regression model's lower performance (Miskell et al., 2015). The feature localized interpretation was implemented to understand the extent of feature contribution, indicating regions where features had positive or negative contributions to on-road pollution, enabling refined street-level pollution control management. The interpretation allows for the identification of significant feature contributions specific to each road segment and aids in prioritizing management efforts accordingly, which allows to comprehensively understand variability in pollutant concentrations and design effective guidelines for urban plannings (Huang et al., 2021).

Our prediction identified localized TRAP hotspots were mainly located in high-rise buildings and congested commercial areas, which aligned with LUR model and the latest fine-scale dispersion model in Hong Kong (Che et al., 2023; Lee et al., 2017). In contrast to previous studies focused on global traffic emission estimations to assess pollution control effects (Matthias et al., 2020), we designed two strategies to assess the impact of improvements in human-intervened traffic transition scenario on reducing TRAP.

At the macro scale, the eliminated congestion yielded limited reductions (5 %-8%). This is because heavy congestion only takes place in 6 % of road segments (and 3 % experienced constant congestion) while 80 % keep good traffic flow (Figure S34). Correspondingly, the causal analysis supplemented prediction results and confirming the effectiveness of human interventions in reducing TRAP. While alleviating traffic congestion may only lead to a limited reduction in pollution, its significance lies in maximizing environmental benefits for localized air quality, particularly during peak hours. Overall, our analysis suggests that increasing the usage of low-emission vehicles and tightening emission standards are essential for achieving broad emission reductions (Wang et al., 2020). However, addressing traffic congestion concurrently can yield greater environmental benefits in areas with high population density.

4.2. *Limitations and future works*

To comprehend the applicability of this study and direct future research, several limitations should be taken into account. Firstly, while the bus platform campaign with repeated routes ensured sample site visitation, it limited the spatial coverage of urban pollution. The sample size is a critical issue, as validating model universality and performance necessitates large-scale datasets, especially in cities with high road density. Reduced model performance in spatial CV indicates that imbalanced spatial distribution of pollutants may hinder the model's spatial prediction capabilities with limited sampling segments. Google View cars have exhibited a random drive-by paradigm that facilitates extensive spatial coverage (Apte et al., 2017). Mobile sensing capabilities can be improved by randomly dispatching vehicle fleets, simultaneously expanding coverage in both spatial and temporal dimensions (O'Keeffe et al., 2019). Secondly, defining the traffic index as a proxy for real traffic intensity status rather than traffic speed may weaken traffic status representation in different road contexts. Considering that bus speed is generally lower than vehicle fleet speed, incorporating large-scale taxi fleets could provide a more reliable road speed measurement and support fine-scale emission inventory estimation in future research. Thirdly, it should be clarified that our model and campaign aim to estimate on-road pollutant variations rather than area surface predictions, even though we have achieved grid-based surface predictions with comparable performance. We assume that a comprehensive urban profile can be best explained through the integration of mobile-based and fixed-site-based models. The two components represent distinct urban pollution patterns with inherently inconsistent driving factors. The on-road pollutants presented highly localized variation that mainly driven by local features, while urban background pollution is driven by localized features and regional influence in combination. As air sensor applications in fixed and mobile networks increase, developing models that fuse these two paradigms could become a significant area of future research. Finally, it should be noted that our scenario analysis for TRAP prediction reduction could be further refined. While the predictive comparative analysis is effective in identifying patterns, it relies on correlation rather than causality, which could also influence 'what-if' pollution levels. Similarly, the causal analysis makes specific assumptions about certain confounding features, which may not fully capture the complexity of real-world traffic and pollution dynamics. To enhance the applicability and robustness of our model, it is crucial to assess model performance across a variety of urban environments and to incorporate environmental changes that may impact air pollution dynamics. This includes considering factors such as meteorological variations, land use patterns, and the influence of secondary pollutants. Thereby providing a more comprehensive and adaptable framework for urban air pollution modeling. Therefore, despite our work offering potential pathways for improving air quality estimations and predicting the impact of targeted policy interventions, further research is required to fully understand the multifaceted relationship between traffic congestion and air pollution.

5. Conclusion

In this study, we built a thorough framework for prediction, analysis, and evaluation with human interventions. A deep learning-based air model was developed to predict fine-scale TRAP spatial distribution, achieving R^2 values of 0.72 for NO and 0.69 for NO_2, demonstrating excellent performance in complex urban environments. Incorporating real-time Google traffic data—a new, effective, and accessible data source—markedly improved model accuracy, underscoring the importance of traffic patterns in estimating TRAP. The interpretable machine learning method revealed the significance of traffic-related features and key contributions of the newly introduced traffic index that facilitated rapid responses to urban infrastructure changes. Recognizing feature interactions highlighted their synergistic effects on pollution, leading to more nuanced and effective pollution control strategies. Meanwhile, the localized interpretation aids in identifying regions where feature contributions vary, allowing for refined street-level pollution management and the design of effective urban planning guidelines. The scenario provides detailed insights into TRAP abatement, highlighting that while alleviating traffic congestion yields limited reductions in pollution, it maximizes environmental benefits for localized air quality during peak hours. The integration of a high-performance model and new data improves prediction accuracy and deepens our understanding of the variables affecting air quality, which is essential for informing future policy and environmental initiatives.

CRediT authorship contribution statement

Peng Wei: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Song Hao:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Yuan Shi:** Writing – original draft, Methodology, Conceptualization. **Abhishek Anand:** Writing – original draft, Methodology. **Ya Wang:** Validation, Resources. **Mengyuan Chu:** Validation, Resources. **Zhi Ning:** Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This study was supported by General Research Fund (GRF) of Hong Kong Special Administrative Region, China (16212022, 2022-2023), Environment and Conservation Fund (ECF/21/2021).

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2024.108992.

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